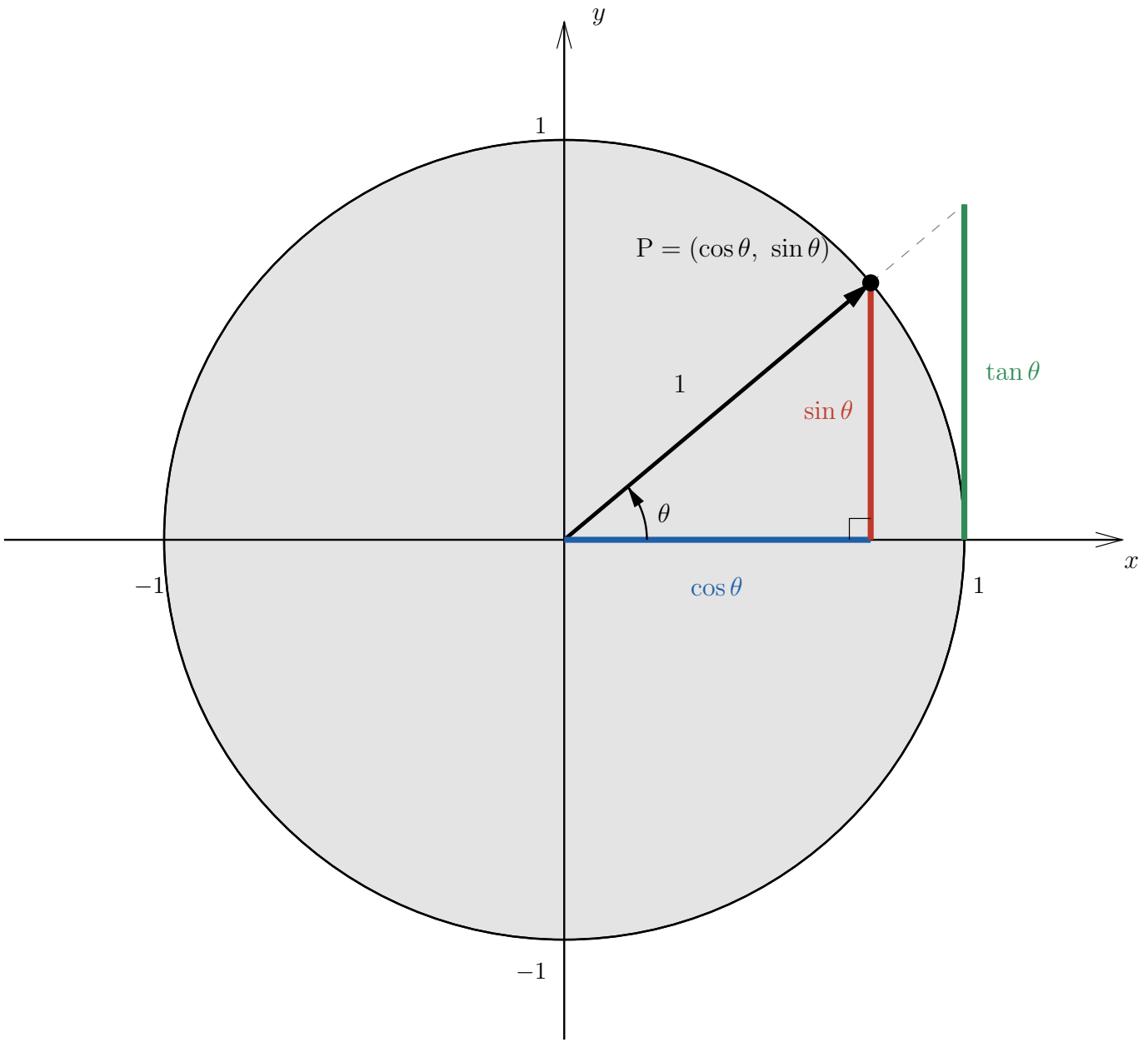
Unit circle and the trigonometric functions

$\sin\theta$, $\cos\theta$ and $\tan\theta$ as lengths of segments, here for $\theta = 40°$



$\sin^2\theta + \cos^2\theta = 1$

$\tan\theta = \sin\theta / \cos\theta$

For $\theta = 40°$:

$\cos\theta = 0.766$

$\sin\theta = 0.643$

$\tan\theta = 0.839$